

AI-Powered Self Drive Car Rental System

PROJECT REPORT

Submitted by

Siva Balan K	230701316
Siva Santhosh R	230701318
Sree Dev S	230701331
Praveen Kaanth D L	230701399

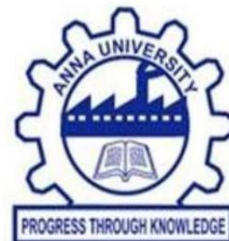
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BONAFIDE CERTIFICATE

Certified that this project report “AGENTIC AI-POWERED SELF DRIVE CAR RENTAL SYSTEM” is the bonafide work of “SIVA BALAN K (230701316), SIVA SANTHOSH R (230701318), SREE DEV S (230701399) and PRAVEEN KAANTH D L (230701399)” who carried out the project work under my supervision. Certified further that to the best of my knowledge the work reported herein does not form part of any other thesis or dissertation on the basis of which a degree or award was conferred on an earlier occasion on this or any other candidates.

This project addresses the following Sustainable Development Goal:
SDG Goal No: 11 Sustainable Cities and communities.

SIGNATURE

Dr. J. Manoranjini
HEAD OF THE DEPARTMENT
Department of Computer Science and
Engineering
Rajalakshmi Engineering College
Chennai - 602 105

SIGNATURE

Dr.S.Anantha Siva Prakasam
SUPERVISOR
Professor
Department of Computer Science and
Engineering
Rajalakshmi Engineering College
Chennai - 602 105

Submitted for the Project Viva - Voce Examination held on
_____.

INTERNAL EXAMINER

EXTERNAL EXAMINER

ABSTRACT

The proposed Car Rental Recommendation and Monitoring System is an Intelligent car Rental platform recommending the most suitable car based on users requirements. The system takes inputs like number of passengers, traveling area or distance, number of members and luggage requirements to identify and recommend appropriate vehicle. Once the user rents and selects the car, the system tracks the vehicle in real time with latitude and longitude coordinates, and the user can see where the car is on a map. The platform also provides key vehicle and service information, including vehicle details, services available.

Besides, a deep learning-based car damage detection module is embedded to detect vehicle damages from images. To be able to train a deep learning model for analysis and detection of visible damages on vehicle images uploaded or captured, we collect a car damage image dataset. The proposed system offers a convenient and technology-driven solution for both customers and car rental service providers by integrating intelligent car recommendation, real-time location tracking, vehicle service information and automated damage detection.

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Siva Balan K

Siva Santhosh R

Sree Dev S

Praveen Kaanth D L

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LIST OF ABBREVIATIONS

ABBREVIATION	EXPANSION
AI	Artificial Intelligence
CNN	Convolutional Neural Network
CSS	Cascading Style Sheets
DFD	Data Flow Diagram
DL	Deep Learning
GAN	Generative Adversarial Network
HTML	Hypertext Markup Language
IDLE	Integrated Development and Learning Environment
RAM	Random Access Memory
UML	Unified Modeling Language

CHAPTER 1

INTRODUCTION

1.1 GENERAL

Car Rental Recommendation and Monitoring System is an intelligent platform to help the process of selection and management of rental vehicles. The system suggests a suitable car on the details provided by the user, like number of persons, traveling area, number of members and luggage requirements. Once the vehicle is selected, the user can trace the rented car using latitude and longitude information and see its present location on a map. Also, the system delivers car- and service-related information and includes a deep learning-based module for detecting vehicle damage from images.

The proposed system integrates intelligent car recommendation, real time location tracking, vehicle service information and automated damage detection to provide a convenient and technology driven solution to the customers as well as the car rental service providers.

1.2 PROBLEM STATEMENT

The process of renting a car currently requires the customers to manually search and compare the vehicles according to their travel needs, number of passengers, luggage capacity and other preferences. This may lead to difficulties for users to quickly find the most appropriate vehicle. Moreover, rental service providers may also have difficulties in continuous vehicle location tracking and vehicle damage detection after or during rental usage. “Manual inspection of a vehicle can be time-consuming, and may not be consistent in identifying damage.

Besides, rental platforms might lack an intelligent recommendation mechanism capable of jointly taking into account multiple customer requirements. Thus, it is necessary to develop an integrated system for car rental that can provide intelligent car recommendation, real-time location tracking,

service information and automated vehicle damage detection using deep learning, making the rental process more efficient, reliable and convenient.

1.3 OBJECTIVES

- To develop an intelligent car rental system which suggests suitable vehicles based on the user's requirements.
- To reduce the issue of manually selecting a rental vehicle based on number of passengers, travel area and luggage needs.
- Real time tracking of position of vehicles using latitude and longitude.
- To develop a deep learning approach to detect car damage automatically from images, assisting rental providers to inspect and monitor vehicle conditions in a more efficient way..

1.4 EXISTING SYSTEM

In the current car rental system, the customers typically choose the vehicles manually depending on the available car details such as size, seating capacity and rental cost. All customer requirements, including number of passengers, traveling area and luggage capacity, may not be considered in the selection process. Also often the estimation of the rental price is based on predefined rates or manual calculations instead of an intelligent prediction model. Vehicle damage inspection is typically done by staff physically inspecting the vehicle manually, which can be time consuming and may depend on the inspector's observation. These limitations call for an intelligent system that could assist in vehicle selection, rental price estimation, and automatic damage detection.

1.4.1 Disadvantages

- Real-time tracking and online rental requires a stable internet connection.

- Latitude and longitude tracking is not always accurate to the location of the vehicle.
- The car damage detection model needs a large and diverse car damage image dataset.

1.5 PROPOSED SYSTEM

The proposed system is an intelligent Car Rental Recommendation and Monitoring System, which aims to help users select a suitable rental car according to their particular travel needs. The system takes inputs like number of persons, traveling area and luggage details. It analyzes these requirements and recommends the most suitable car. Once the user selects a car, the system gives real time tracking by using the latitude and longitude information of the vehicle so that the current car location can be displayed. The system also provides relevant car and service details so that the user can understand the available vehicle services and rental information.

Also included is a deep learning based module for detecting car damages. To train a deep learning model, we collect a dataset of images of various types of car damage. For example: given a picture of a car, the trained model looks at the picture and finds that there is visible damage. This feature can aid rental service providers in maintaining proper vehicle records and inspecting vehicles before and after rental usage. Thus, the proposed system integrates car recommendation, real-time vehicle tracking, car service information and automated damage detection in a single intelligent car rental platform.

1.5.1 Advantages

- Recommends appropriate cars according to number of persons, traveling area, members and luggage requirements.
- Saves users time when manually comparing different rental cars.

- Users can track the current location of the rented car in latitude and longitude.

1.6 SCOPE OF THE PROJECT

The project scope includes intelligent car recommendation, car tracking, car service information and automated car damage detection. Passengers can list their travel and passenger needs and receive recommendations for the appropriate car. You can follow the selected vehicle by its latitude and longitude coordinates. The system is also able to retain and display pertinent vehicle and service information. Vehicle inspection: We use the car damage image dataset to train a deep learning model to detect visible damage from car images. The system can be useful for both the customers and the car rental service providers.

1.7 MOTIVATION OF THE PROJECT

The motivation of this project is to make car rental process more convenient, intelligent and reliable for both customers and rental service providers. It can be difficult for users to decide which vehicle is best for their traveling needs, number of passengers and luggage capacity. Likewise, manually tracking vehicle locations and inspecting vehicles for damage may require extra work. The proposed system integrates car recommendation, location tracking, service information and deep learning-based damage detection into one platform, which can reduce manual works and provides a more efficient car rental experience .

CHAPTER 2

LITERATURE SURVEY

2.1 Literature Survey

[1] Mallios et al. (2025) address the evaluation of vehicle damages in insurance operations that usually require costly and time-consuming manual inspection. The authors propose a pipeline that accepts as input photographs of damaged vehicles taken from multiple angles and structured vehicle information and estimates the damage severity. Then, the damage cost estimation is performed using the extracted features and the extent of damage is determined by applying semantic segmentation for the damaged vehicle parts. The model is trained and tested on a large dataset of historical insurance claims captured using mobile-phone hardware.

[2] Kyu and Woraratpanya (2026) work on the assessment of realistic car damage images and the limited availability of real-world vehicle damage datasets. We employ Generative Adversarial Networks (GANs) to generate realistic images of vehicle damage. We suggest Advanced Mahalanobis Distance Learning (AMDL) to measure the realism of each generated image. The proposed method exploits the deep feature embeddings by the pre-trained CNNs with the adaptive regularization and the pseudo-inverse refinement. The generated car damage images are also validated by CNN and Vision Transformer models in a classifier based manner.

[3] Loureiro et al. (2026) examine the automated vehicle damage detection and vehicle-parts segmentation for shared vehicle systems, where the manual inspection can lower the operational efficiency. The authors propose SYNDCAR, a synthetic dataset based on generative AI that contains realistic vehicle damage images annotated for damage detection and vehicle-parts segmentation. We use hyperparameter tuning and transfer learning to evaluate CNN- and Transformer-based models. Experimental

results demonstrate that YOLOv8-X has high accuracy and YOLOv12-S strikes a good balance between accuracy and computational efficiency.

[4] Liu and Wang (2024) proposed a method for vehicle path tracking using optimal control. We develop a nonlinear 4-degree-of-freedom vehicle model and formulate the path tracking problem to a nonlinear programming problem by discretizing the control and state variables with the local collocation method. Additionally, an efficient approach for computing partial derivatives is presented to increase computational efficiency. Furthermore, real-vehicle experiments confirm the effectiveness of the proposed methodology for vehicle path tracking.

[5] Wu et al. (2024) propose a Fast Model Predictive Control (FMPC) method for vehicle path tracking aiming at enhancing real-time performance in highly nonlinear vehicle systems. Vehicle behavior is modeled using a recurrent neural network with symmetric saturating linear transfer functions. To realize faster online computation, the predictive control problem is formulated as a mixed-integer linear programming problem. The CarSim/Simulink simulations show that the tracking accuracy and the controller solution efficiency are improved compared with the conventional RNN and LSTM approaches.

2.1 COMPARISON AND DISCUSSION

Table 2.1 Comparison of Related Work

WORK	FOCUS	RELEVANCE TO THIS PROJECT
Mallios et al. [1]	Damage severity estimation from mobile images using segmentation	Supports automated image-based damage assessment
Kyu and Woraratpanya [2]	Realism assessment of GAN-generated car damage images	Addresses limited availability of car damage datasets
Loureiro et al. [3]	Synthetic dataset for damage detection and parts segmentation	Supports damage detection for vehicle inspection
Liu and Wang [4]	Optimal control for vehicle path tracking	Background on vehicle tracking and monitoring
Wu et al. [5]	Fast predictive control for vehicle path tracking	Background on vehicle tracking and monitoring

The surveyed work shows that automated damage assessment from images is an active research area and that the availability of a sufficiently large and diverse damage dataset strongly influences the quality of detection. The vehicle path tracking studies [4], [5] concern the control of vehicle motion; they provide background on vehicle monitoring, whereas the present project tracks a rented vehicle through its latitude and longitude coordinates.

2.2 CONCLUSION

The literature shows that image-based damage detection and vehicle tracking have been studied mainly as independent problems. The proposed system integrates car recommendation based on user requirements, location tracking, service information and deep learning based damage detection in a single car rental platform.

CHAPTER 3

SYSTEM DESIGN

3.1 SYSTEM ARCHITECTURE

The proposed system architecture is based on two functional paths. In the damage detection path, the car damage dataset is preprocessed and features are extracted. Then, the Inception V3 algorithm is used to build the model. The administrator provides images of cars as input to the model, which then analyzes them and reports any damage found. In the recommendation path, the user enters passenger data, the system recommends suitable cars, the user selects a car and its track location, and car details are displayed.

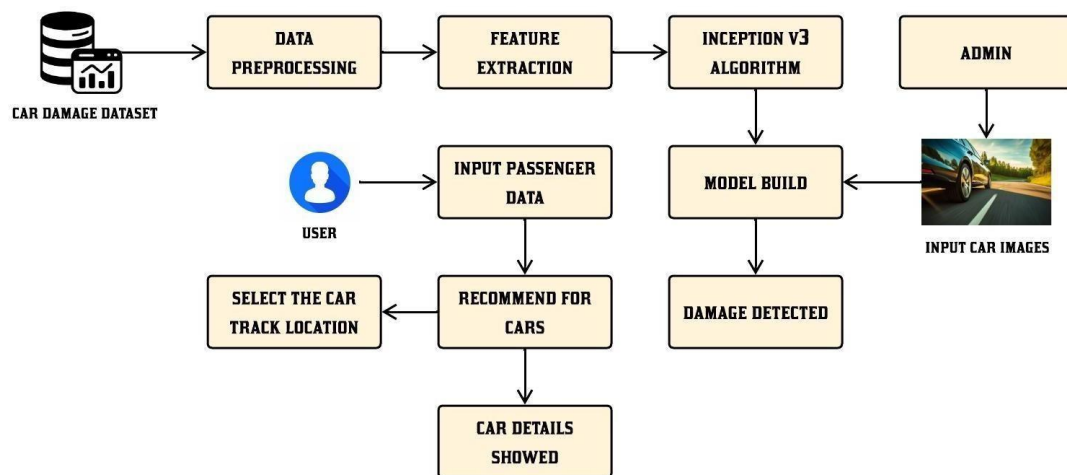


Figure 3.1 System Architecture

3.2 SYSTEM REQUIREMENTS

3.2.1 Software Requirements

Table 3.1 Software Requirements

COMPONENT	TECHNOLOGY / REQUIREMENT
Operating System	Windows 10/11
Front End	HTML, CSS
Scripts	Python language
Tool	Python IDLE

3.2.2 Hardware Requirements

Table 3.2 Hardware Requirements

RESOURCE	CONFIGURATION
Processor	Intel
Speed	1.1 GHz
RAM	8 GB
Hard Disk	500 GB

3.3 UML DIAGRAMS

3.3.1 Use Case Diagram

The use case diagram illustrates the interaction of the user and the administrator with the system. The user enters passenger data, receives car recommendations, and tracks the selected car. The admin manages the car damage dataset, preprocessing, feature extraction, model creation, and inputting car images for damage detection.

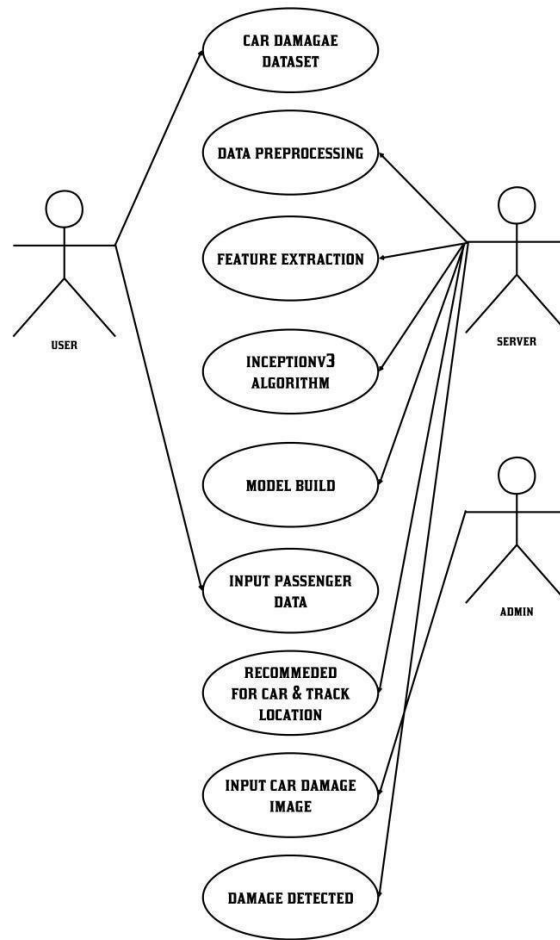


Figure 3.2 Use Case Diagram

3.3.2 Activity Diagram

The activity diagram shows the sequence of activities in the system beginning from the car damage dataset, preprocessing of data, feature extraction, Inception V3 algorithm and model build, input of passenger data, car recommendation, input of car damage image and damage detection.

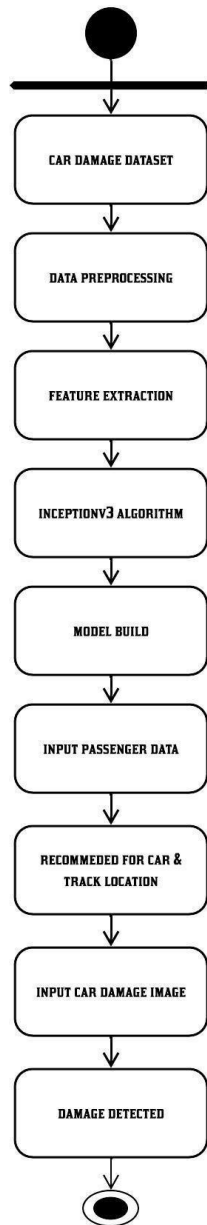


Figure 3.3 Activity Diagram

3.3.3 Data Flow Diagram

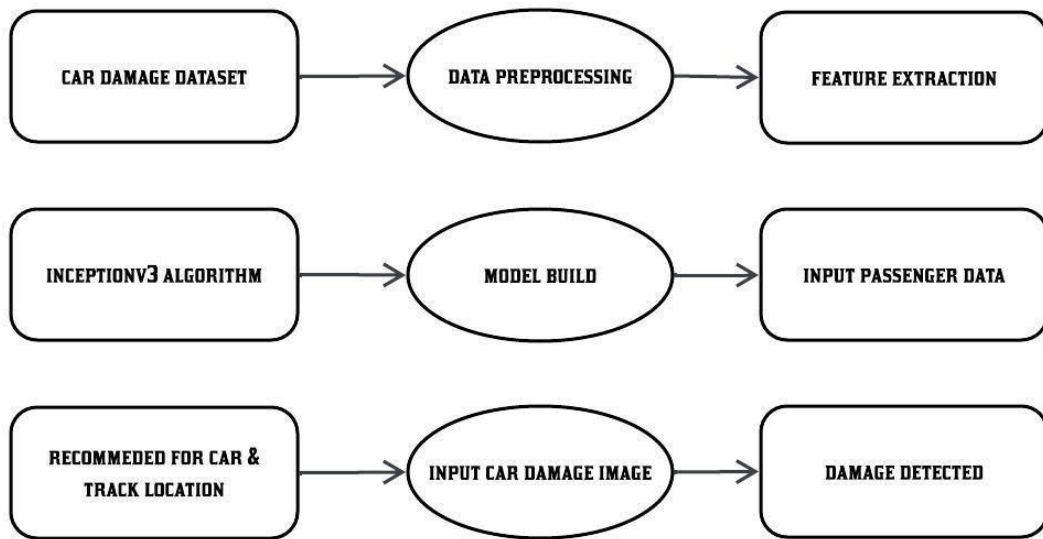


Figure 3.4 Data Flow Diagram - Level 0

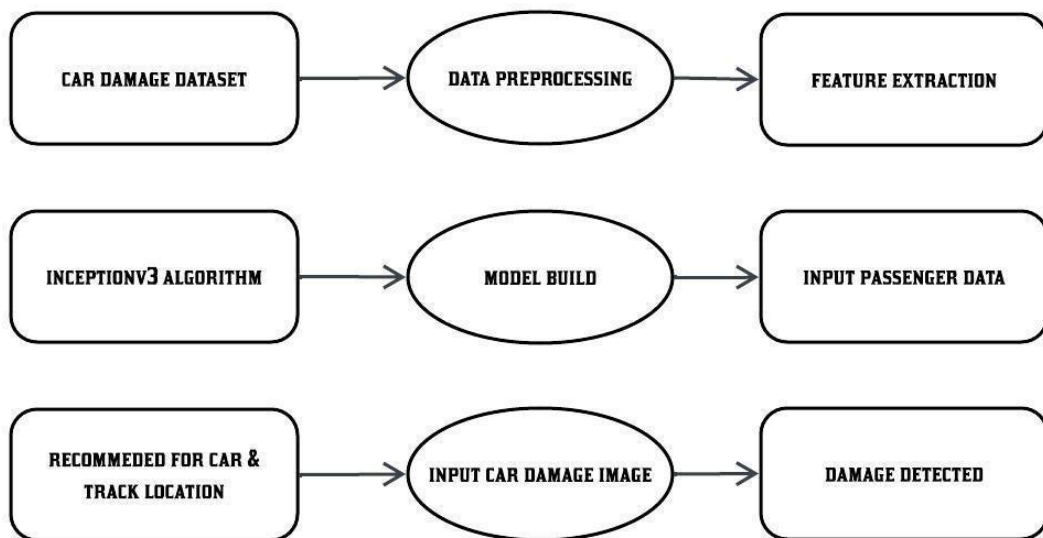


Figure 3.5 Data Flow Diagram - Level 1

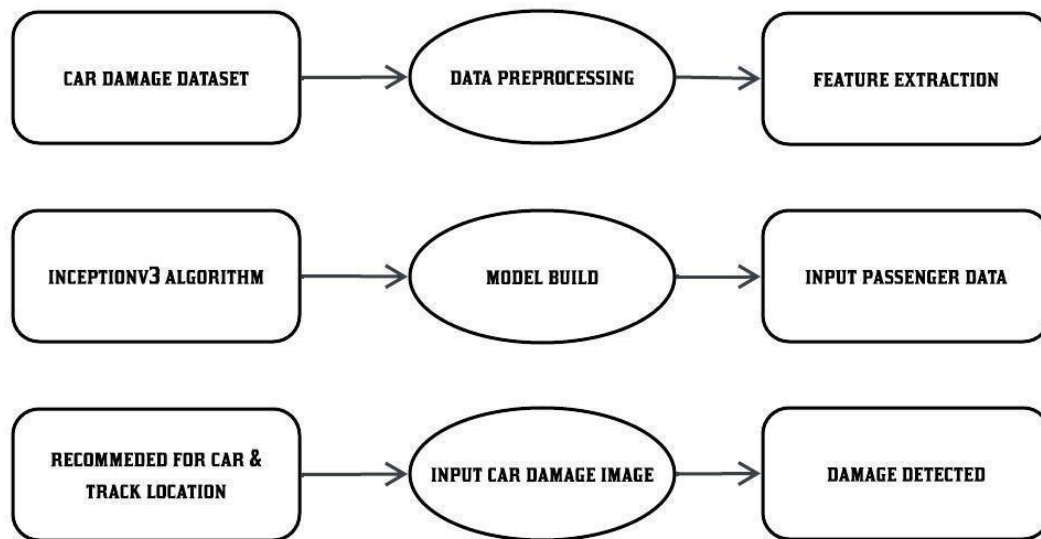


Figure 3.6 Data Flow Diagram - Level 2

3.3.4 Overall Diagram

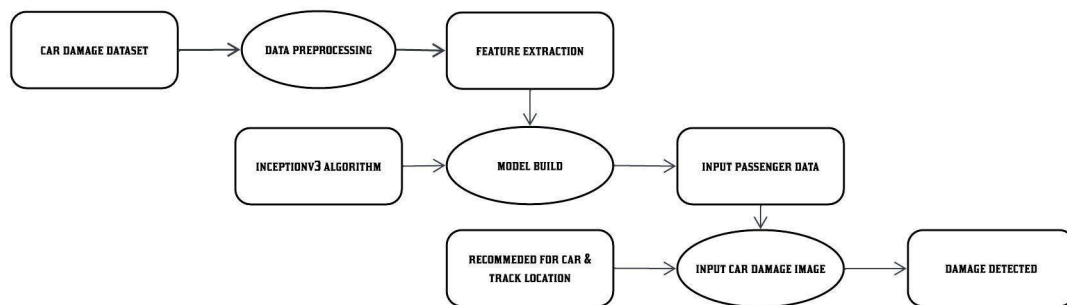


Figure 3.7 Overall Diagram

3.4 FEASIBILITY STUDY

In this phase analysis of the feasibility of the project is done so that the proposed system does not become a burden to the organization. To properly perform the feasibility analysis, it is important to understand the major requirements of the system.

3.4.1 Economical Feasibility

This study is conducted to verify the system's economic impact on the organization. The company has a limited amount of money to spend on the research and development of the system and the expenditures have to be justified. The developed system fits well within the budget as most of the technologies used are freely available and only the customized products had to

be purchased.

3.4.2 Technical Feasibility

The system's technical requirements are verified in this study. Any system developed should not be a high demand on the available technical resources, as this would lead to high demands being placed on the client. The implementation of the developed system requires modest requirements as only minimal or no changes are required.

3.4.3 Social Feasibility

This part of research is to check the level of acceptance of system by the user which includes the process of training the user to use the system efficiently. The system should not be a threat to the user, but the user should accept it as a necessity. The degree of acceptance depends on the way of educating the user about the system and the way of making the user familiar with the system so that the user can give constructive criticism as the final user of the system.

CHAPTER 4

PROJECT DESCRIPTION

4.1 SYSTEM MODULES

The system consists of the following modules:

- User Requirement Module
- Car Recommendation and Selection Module
- Car Tracking and Location Module
- Car Service Details Module
- Deep Learning-Based Car Damage Detection Module

4.1.1 User Requirement Module

The module collects the user's requirements for car rental, such as number of persons, number of members, area of travel, and luggage information. These inputs are used to understand the travel needs of the user and what type of vehicle is required

4.1.2 Car Recommendation and Selection Module

This module analyzes the user's requirements and recommends a suitable car based on the passenger capacity, traveling area and luggage space. The user can see the recommended vehicles and choose the selected car for rent.

4.1.3 Car Tracking and Location Module

This module tracks the selected rental car using latitude and longitude coordinates. The location data is used to show the current location of the vehicle on the map which helps the users and rental providers to track the vehicle during the rental period.

4.1.4 Car Service Details Module

This module gives information about the selected car and the services that are available for it. Users can check the vehicle information and service related information to know about the features and facilities provided with the rental car.

4.1.5 Deep Learning-Based Car Damage Detection Module

This module uses a deep learning model to detect visible damage of the car from vehicle images. The model is trained on a data set of different pictures of car damage. Once trained the system inspects a given car image and determines if any visible damage is present, assisting in vehicle inspection and rental management.

4.2 PROPOSED SYSTEM ALGORITHM

4.2.1 Inception V3 Algorithm

Inception V3 is similar to and has all the features of Inception V2 with the following changes and additions:

- RMSprop optimizer.
- Batch normalization of the auxiliary classifier fully connected layer.
- Use of 7x7 factorized convolution.
- Label smoothing regularization: the technique to regularize the classifier by estimating the effect of label dropout during training. It prevents the classifier from being too confident on a class prediction. We get 0.2% improvement in the error rate by adding label smoothing.

The Inception V2 architecture replaces the 5x5 convolution with two 3x3 convolutions. This reduces computation time and thus increases speed of computation as a 5x5 convolution is 2.78 times more costly than a 3x3 convolution. The performance of the architecture is therefore increased by using two 3x3 layers instead of one 5x5 layer.

4.3 TECHNOLOGIES USED

4.3.1 Python

Python is a high level , object oriented , general purpose programming language created by Guido van Rossum . Many domains use it, including web design, software development, game development, machine learning and artificial intelligence, and data analytics.

It has a simple syntax similar to the English language , so you can write programs with fewer lines of code . Also , it has open source libraries that increase the productivity of developers .

4.3.2 Python Standard Library and Packages

The Python standard library is large, and contains built-in modules for things like file I/O, and Python-written modules that provide standard solutions to common programming problems. Along with the standard library, thousands of third-party components are available from the Python Package Index and are installed and imported into programs as packages.

4.3.3 Front End

On the front end, the system is developed using HTML and CSS. The scripting is done in Python language where the development tool used is Python IDLE.

CHAPTER 5

IMPLEMENTATION AND RESULT DISCUSSION

5.1 IMPLEMENTATION

The proposed AI-Based Smart Car Rental Recommendation, Real-Time Monitoring and Damage Detection System is developed as a single web-based platform with Python as backend and HTML/CSS as frontend. The system is implemented in separate modules: user requirement collection, recommendation of suitable cars, vehicle information management, tracking rented vehicles, display of services for cars and detection of vehicle damage using deep learning.

The User Requirement Module collects important details from the customer such as number of passengers or members, traveling area or distance and luggage requirements. These inputs are processed and fed to the recommendation module which determines the vehicles that fit the customer's travel needs.

The Car Recommendation and Selection Module compares the user requirements with the available vehicle information. The system filters the appropriate vehicles according to passenger capacity, travel requirements and luggage capacity and shows suitable car recommendations. The customer can then choose a vehicle to be rented. Car Tracking and Location Module: With Car Tracking and Location Module you can monitor the rented vehicle in real time using latitude and longitude coordinates. The location information is processed by the system and displayed on a map, allowing the customer or rental service provider to view the current location of the vehicle.

The Car Service Details Module maintains and displays important information about the selected vehicle and its available services. This includes relevant vehicle details and rental-related information, helping customers understand the features and services associated with the selected car.

The Car Damage Detection Module uses a deep learning approach to identify visible vehicle damage from images. The car damage image dataset is first collected and preprocessed before being used for model development. The implementation uses the Inception V3 algorithm for feature extraction and image analysis. The trained model receives a car image as input and analyzes the visual features to identify whether visible damage is present.

5.2 SOFTWARE TESTING

5.2.1 Unit Testing

Unit testing is the creation of test cases that verify the internal program logic is functioning correctly and that program inputs produce valid outputs. All decision branches and internal code flows should be verified. It is test of individual software components of the application. It is performed after completion of an individual component before integration. It is structural testing based on knowledge of its construction. Unit tests are the basic tests at component level and test a specific business process, application and/or system configuration. They make sure that each unique path of a business process is working properly according to the documented specifications and has well-defined inputs and expected outcomes.

5.2.2 Integration Testing

Integration tests are designed to test integrated software components to determine whether they run as one program. Testing is event driven and is more concerned with the basic outcome of screens or fields. Integration tests demonstrate that, although the components were individually satisfactory as shown by unit testing, the combination of components is correct and consistent. Integration testing is specifically aimed at exposing the problems that arise from the combination of components.

5.2.3 Functional Testing

Functional tests provide systematic demonstrations that the functions tested are available as specified by the business and technical requirements, system documentation and user manuals. Functional testing is centered on the following items:

- Valid input: Identified classes of valid input should be accepted.
- Invalid input: Identified classes of invalid input should be rejected.
- Functions: Identified functions should be exercised.
- Output: Identified classes of application outputs should be exercised.

Systems/procedures: Interfacing systems or procedures should be invoked.

Organization and preparation of functional tests is based on requirements, key functions or special cases of tests. Systematic coverage of the identified business process flows, data fields, predefined processes and successive processes should also be considered for testing. Additional tests are identified before functional testing is completed and the effective value of current tests is determined.

5.2.4 System Testing

System testing ensures that the entire integrated software system meets requirements. It tests a configuration to ensure known and predictable results. An example of system testing is the configuration-oriented system integration test. System testing is based on process descriptions and flows, emphasizing pre-driven process links and integration points.

5.2.5 White Box Testing

White box testing is testing in which the software tester has knowledge of the inner workings, structure and language of the software, or at least its purpose. It is used to test areas that cannot be reached from a black box level.

5.2.6 Black Box Testing

Black box testing is testing the software without having any knowledge of the inner workings, structure or language of the module that is being tested. Like other types of testing, black box tests need to be written from a definitive source document such as a specification or requirements document. The software being tested is treated as a black box which you cannot see inside. The test supplies inputs and reacts to outputs without considering how the software operates.

5.1.7 Model Performance Evaluation

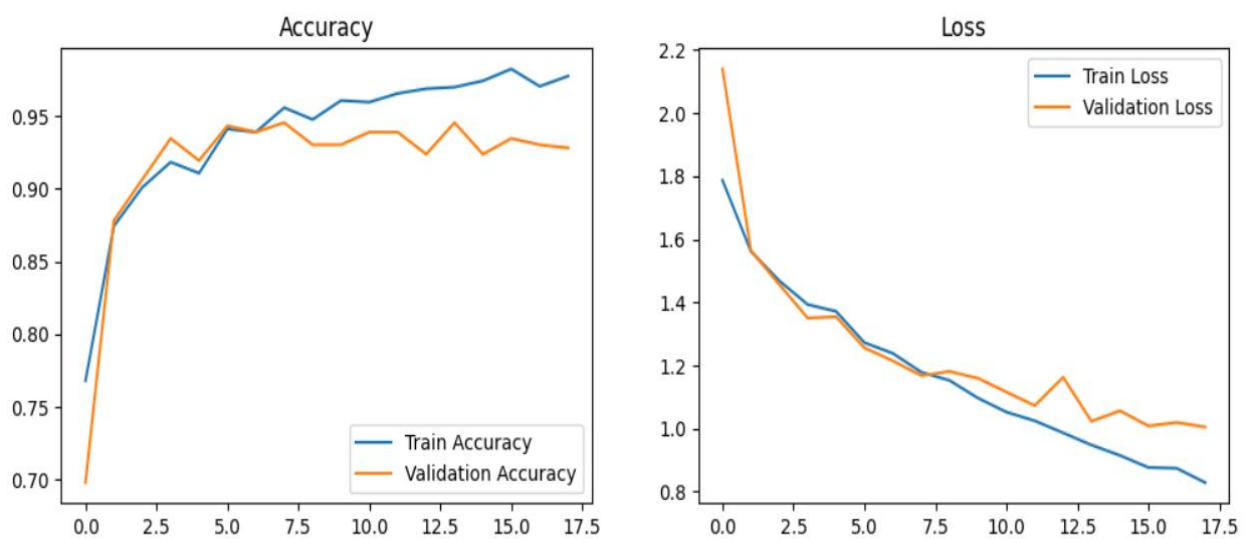


Figure 5.1 Train Accuracy and Validation Accuracy

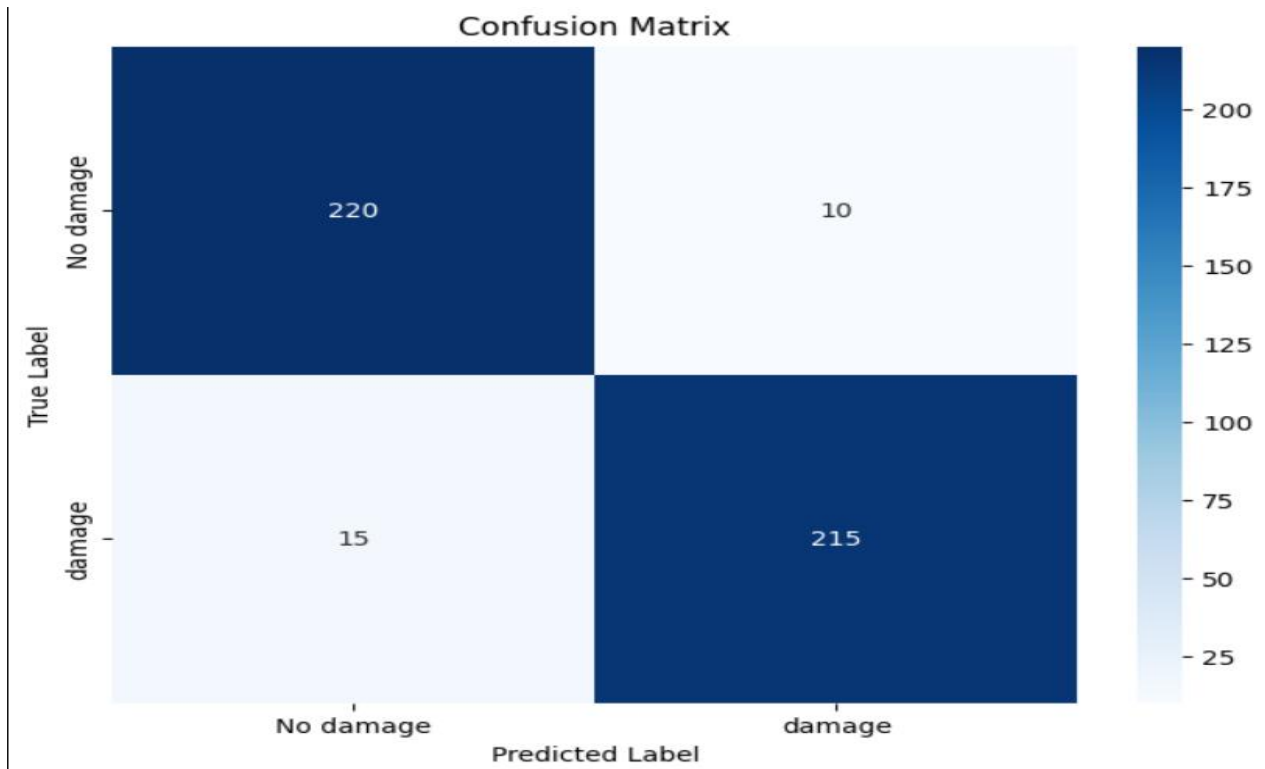


Figure 5.2 Confusion Matrix

5.3 RESULTS

The output screens of the system are shown below. The screenshots are to be inserted in the reserved areas.

The screenshot shows the DriveX Car Rental website. The header includes the logo, navigation links (HOME, ADMIN, USER, VEHICLES, DEALS, SERVICES, ABOUT US, CONTACT US), and a 'BOOK NOW' button. The main banner features the text 'Find The Perfect Car For Every Journey' and a 'BOOK YOUR RIDE' button. Below the banner, there are tabs for 'Rent a Car', 'Long Term Rental', and 'Corporate Rental'. The 'Rent a Car' tab is active, showing a form with the following fields: 'PICK-UP LOCATION' (Enter city or airport), 'PICK-UP DATE' (May 24, 2024), 'PICK-UP TIME' (10:00 AM), and 'DROP-OFF DATE' (May 27, 2024). A 'SEARCH CARS' button is located to the right of the date fields.

Figure 5.3 User Requirement Input Screen

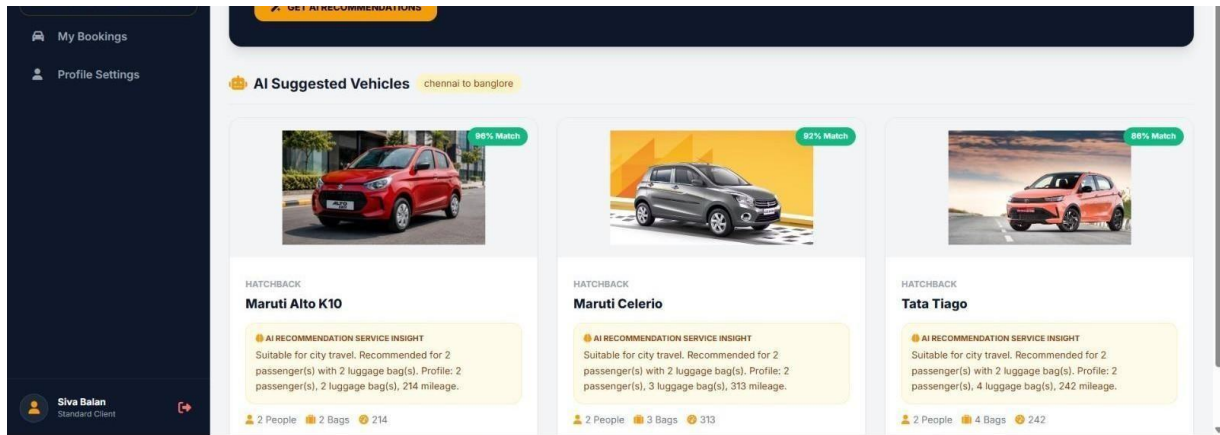


Figure 5.4 Car Recommendation Output

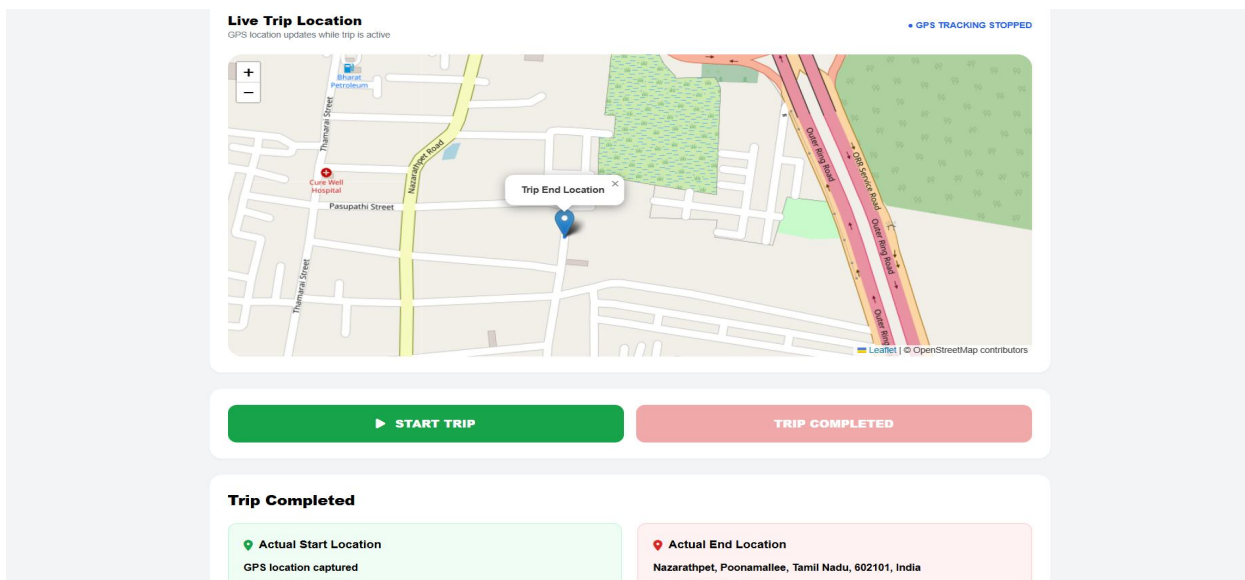


Figure 5.5 Car Tracking and Location Map

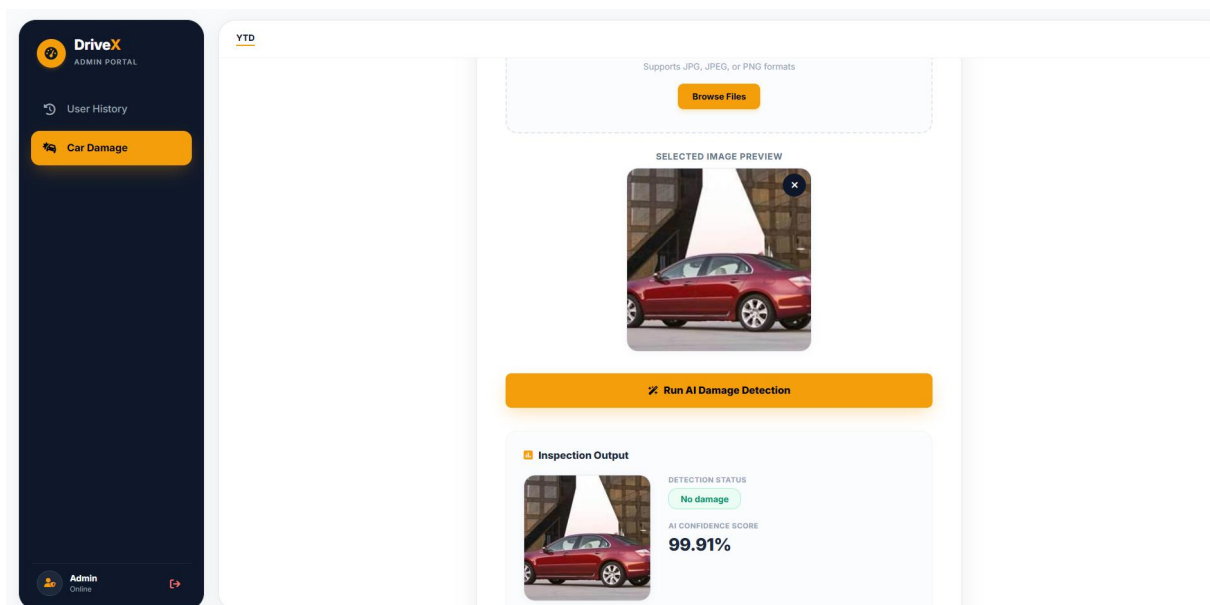


Figure 5.6 Car Damage Detection Output

CHAPTER 6

CONCLUSION AND FUTURE WORK

6.1 CONCLUSION

The proposed Car Rental Recommendation and Monitoring System provides an integrated solution for intelligent vehicle selection, tracking, service information and vehicle damage detection. The system recommends cars based on the requirements of the user and also provides tracking of the selected car using latitude and longitude data. The deep learning damage detection module can evaluate images of cars and identify visible damage on the vehicles. Generally, the system offers a technology-based solution to facilitate ease of car rental management and make vehicle monitoring and inspection easier.

6.2 FUTURE ENHANCEMENTS

- Use advanced AI techniques to provide more personalized vehicle recommendations based on user preferences, previous rental history, budget, travel distance and destination.
- Enhance the location tracking module with real-time route monitoring, geofencing and automatic notifications for unusual vehicle movements.
- Expand the damage detection module to detect different types and levels of vehicle damage from multiple images or video inputs.
- Integrate predictive maintenance to identify potential vehicle maintenance needs before major failures occur.
- Add AI-based pricing prediction, online payment and booking integration, voice-based assistance and multilingual support.
- Add driver behavior analysis and a chatbot to help customers with vehicle selection, booking and rental related queries.

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